

```

import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q01.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q01.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    ccd = cv2.GaussianBlur(gray, (5,5), 0)
    cmos = cv2.GaussianBlur(gray, (3,3), 2)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(15, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(ccd, cmap='gray')
    plt.title("CCD Image (GaussianBlur 5x5)")
    plt.axis('off')

    plt.subplot(1, 3, 3)
    plt.imshow(cmos, cmap='gray')
    plt.title("CMOS Image (GaussianBlur 3x3)")
    plt.axis('off')

    plt.show()

# cv2.imshow and cv2.waitKey are typically for local GUI applications,
# and do not work in Colab notebooks.

```

Original Image



CCD Image (GaussianBlur 5x5)



CMOS Image (GaussianBlur 3x3)



```

import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q02.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q02.jpg' is in /content/.")
else:
    # Gaussian blur for sharpening
    blur = cv2.GaussianBlur(img, (7,7), 0)

    # Sharpening using addWeighted
    sharp = cv2.addWeighted(img, 1.5, blur, -0.5, 0)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(cv2.cvtColor(sharp, cv2.COLOR_BGR2RGB))
    plt.title("Improved Image (Sharpened)")
    plt.axis('off')

```

```
plt.show()
```

```
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```

Original Image



Improved Image (Sharpened)



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q03.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q03.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    edges = cv2.Canny(gray, 100, 200)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(edges, cmap='gray')
    plt.title("Detected Objects (Canny Edges)")
    plt.axis('off')

    plt.show()

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```

Original Image



Detected Objects (Canny Edges)



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q04.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q04.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    enhanced = cv2.equalizeHist(gray)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
plt.title("Original Satellite Image")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(enhanced, cmap='gray')
plt.title("Enhanced Image")
plt.axis('off')

plt.show()

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```

Original Satellite Image



Enhanced Image



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q05.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q05.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply binary thresholding
    _, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(binary, cmap='gray')
    plt.title("Binary Image")
    plt.axis('off')

    plt.show()

# cv2.imshow and cv2.waitKey are typically for local GUI applications,
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```

Original Image



Binary Image



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q06.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q06.jpg' is in /content/.")
else:
    small = cv2.resize(img, (300,300))
    gray = cv2.cvtColor(small, cv2.COLOR_BGR2GRAY)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(15, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(cv2.cvtColor(small, cv2.COLOR_BGR2RGB))
    plt.title("Reduced Resolution")
    plt.axis('off')

    plt.subplot(1, 3, 3)
    plt.imshow(gray, cmap='gray')
    plt.title("Grayscale Image")
    plt.axis('off')

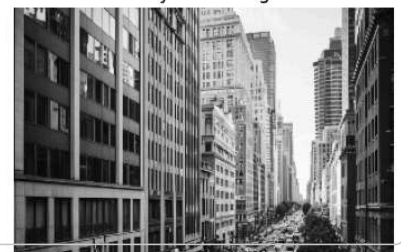
    plt.show()

# cv2.imshow and cv2.waitKey are typically for local GUI applications,
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```

Reduced Resolution

Grayscale Image

Original Image



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
```

```

img = cv2.imread('/content/Q07.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q07.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    enhanced = cv2.equalizeHist(gray)

# Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
plt.title("Original Image")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(enhanced, cmap='gray')
plt.title("Contrast Enhanced")
plt.axis('off')

plt.show()

# cv2.imshow and cv2.waitKey are typically for local GUI applications,
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```

Original Image



Contrast Enhanced



```

import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q08.jpg', 0) # Read as grayscale directly

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q08.jpg' is in /content/.")
else:
    quantized = (img // 32) * 32

# Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Medical Image")
plt.axis('off')

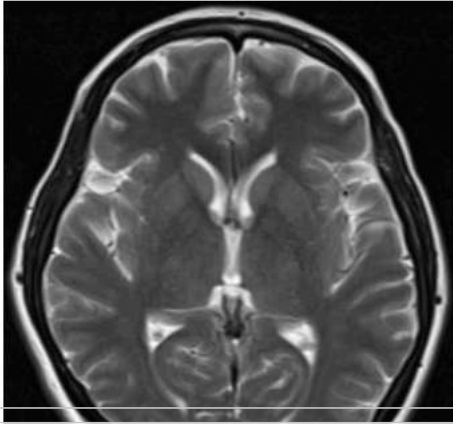
plt.subplot(1, 2, 2)
plt.imshow(quantized, cmap='gray')
plt.title("Quantized Image")
plt.axis('off')

plt.show()

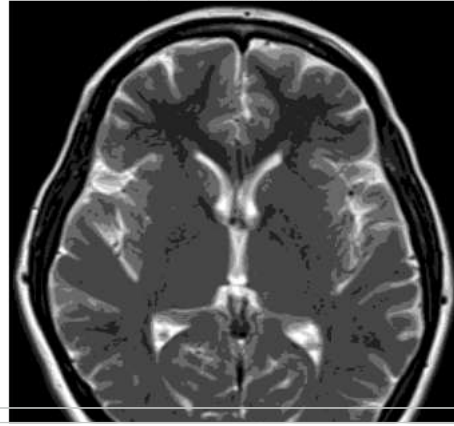
# cv2.imshow and cv2.waitKey are typically for local GUI applications,
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```


Original Medical Image



Quantized Image



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab
img = cv2.imread('/content/Q09.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'Q09.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    edges = cv2.Canny(gray, 50, 150)

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
    plt.title("Original Image")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(edges, cmap='gray')
    plt.title("Lane Detection (Canny Edges)")
    plt.axis('off')

    plt.show()

# cv2.imshow and cv2.waitKey are typically for local GUI applications,
# and do not work in Colab notebooks.
```

Original Image



Lane Detection (Canny Edges)



```
import cv2
import matplotlib.pyplot as plt

# Using the correct path for the image uploaded to Colab (assuming q10.jpg exists)
img = cv2.imread('/content/q10.jpg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not loaded. Please ensure 'q10.jpg' is in /content/.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    details = cv2.Laplacian(gray, cv2.CV_64F) # cv2.CV_64F to allow negative values from Laplacian

    # Using Matplotlib to display images in Colab, as cv2.imshow does not work directly
    plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for matplotlib
plt.title("Original Image")
plt.axis('off')

plt.subplot(1, 2, 2)
# For displaying Laplacian output, often it's helpful to normalize or scale it for visualization
# Since it's CV_64F, it can have negative values. Normalize to 0-255 for display.
plt.imshow(details, cmap='gray', vmin=details.min(), vmax=details.max())
plt.title("Fine Details (Laplacian)")
plt.axis('off')

plt.show()

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```

Original Image



Fine Details (Laplacian)

